



CURRICULUM DOCUMENT

COMPUTER SCIENCE

SAMPOERNA UNIVERSITY
CURRICULUM TASK FORCE | JAKARTA

CURRICULUM DOCUMENT

Computer Science Curriculum Task Force

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**KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
UNIVERSITAS SAMPOERNA
2024**

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FOREWORD

We thank The Almighty God because it is by His grace that the Computer Science (CS) Curriculum Task Force which compiled the Curriculum of the Computer Science Study Program Sampoerna University have completed their task of evaluating the 2021 CS curriculum and updating it to the 2024 CS curriculum.

Evaluation of the 2021 curriculum is carried out by taking into account several important references such as the National Curriculum based on KKNI Competency level 6, the APTIKOM Curriculum Guides, the Independent Learning policy (MBKM) from *Kementerian Pendidikan, Kebudayaan, Riset dan Teknologi* Republic of Indonesia, the Industrial Revolution 4.0 trend, as well as aspects of the vision and mission of the University of Sampoerna and its Faculty of Engineering and Technology. In addition, surveys were also conducted using questionnaires and FGD (Focus Group Discussion) with stakeholders, internally and externally, to obtain inputs which are expected to have a good impact on the development and operation of the study program in the future.

Based on the results of these evaluations, the 2024 Curriculum of the Computer Science Study Program – Sampoerna University, was compiled. The 2024 curriculum is expected to answer the needs and challenges in the Industry 4.0 and digital era where Computer Science graduates are required to contribute positively to the computing world and society in general.

We, the Computer Science Curriculum Task Force, realize that there might be errors and mistakes in this 2024 Curriculum document still. We look forward to constructive comments and suggestions for better improvements to the CS curriculum in the future.

Jakarta, 14 August 2024

On behalf of the Computer Science Curriculum Task Force



(M. Agni Catur Bhakti)

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IDENTITAS PROGRAM STUDI

1	Nama Perguruan Tinggi (PT)	Universitas Sampoerna ☐ PTN ☒ PTS
2	Fakultas	Fakultas Teknik dan Teknologi
3	Jurusan/Departemen	-
4	Program Studi	Ilmu Komputer
5	Status Akreditasi	Baik Sekali, SK LAM INFOKOM Nomor: 022/SK/LAM- INFOKOM/Ak/S/III/2023 tanggal 29 Maret 2023
6	Jumlah Mahasiswa	80
7	Jumlah Dosen	7
8	Alamat Prodi	Jl. Raya Pasar Minggu, Kav 16, Jakarta Selatan
9	Telpon	+62-21-5022-2234, Ext:7718
10	Web Prodi/PT	https://www.sampoernauniversity.ac.id/

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1. Curriculum Foundation

The undergraduate Computer Science Study Program is one of the higher educations in computing and has an important role in solving complex problems in society. The expertise or competencies that are expected from the graduates are the skills in performing computing analysis, which are needed to solve the complex problems. Included also in the competencies of the graduates are the skills to communicate effectively, to collaborate in teamwork, to uphold ethics, norms, standards, and rules in society. Computer Science knowledge is supported with a strong foundation from mathematics and statistics. The foundation from mathematics and statistics is needed in solving complex problems in society.

The role of Computer Science Graduates in this technological advance era of Industrial Revolution 4.0 becomes increasingly challenged. To improve the GCII (Global Competitiveness Index Indonesia), Indonesian government has announced the “Roadmap Making Indonesia 4.0”. The graduates of Computer Science are expected to play a key role in realizing this roadmap.

In answering the above-mentioned challenges, the Undergraduate Curriculum of Computer Science (CS) Study Program at Sampoerna University (SU) must be evaluated regularly and revised when necessary to ensure the quality of the graduates that can fulfill the demands in the industries and societies. This is the motivation in continuously improving the Curriculum of CS Study Program.

In evaluating and revising the curriculum, below are the rules, laws, and the policy that are used as the reference:

1. Undang-Undang Republik Indonesia Nomor 14 Tahun 2005 tentang Guru dan Dosen (Lembaran Negara Republik Indonesia Tahun 2005 Nomor 157, Tambahan Lembaran Negara Republik Indonesia Nomor 4586);
2. Undang-Undang Republik Indonesia Nomor 12 Tahun 2012 tentang Pendidikan Tinggi (Lembaran Negara Republik Indonesia Tahun 2012 Nomor 158, Tambahan Lembaran Negara Republik Indonesia Nomor 5336);
3. Peraturan Presiden Republik Indonesia Nomor 8 Tahun 2012, tentang Kerangka Kualifikasi Nasional Indonesia (KKNI);
4. Peraturan Menteri Pendidikan dan Kebudayaan Republik Indonesia Nomor 73 Tahun 2013, Tentang Penerapan KKNI Bidang Perguruan Tinggi;
5. Peraturan Menteri Riset, Teknologi, dan Pendidikan Tinggi Nomor 59 tahun 2018, tentang Ijazah, Sertifikat Kompetensi, Sertifikat Profesi, Gelar dan Tata Cara Penulisan Gelar di Perguruan Tinggi;
6. Keputusan Menteri Riset, Teknologi, dan Pendidikan Tinggi No. 123 Tahun 2019 tentang Magang dan Pengakuan Satuan Kredit Semester Magang Industri untuk Program Sarjana dan Sarjana Terapan;

7. Peraturan Menteri Pendidikan dan Kebudayaan No. 7 Tahun 2020 tentang Pendirian Perubahan, Pembubaran Perguruan Tinggi Negeri, dan Pendirian, Perubahan, Pencabutan Izin Perguruan Tinggi Swasta;
8. Peraturan Menteri Pendidikan dan Kebudayaan No. 22 tahun 2020, tentang Rencana Strategis Kementerian Pendidikan dan Kebudayaan;
9. Peraturan Menteri Pendidikan dan Kebudayaan No. 53 tahun 2023, tentang Penjaminan Mutu Pendidikan Tinggi.

To develop the substantial aspect of the curriculum, the following references are used:

1. APTIKOM OBE/KKNI/SKKNi Curriculum Guides 2022 - 2024.
2. Bloom Taxonomy.
3. Previous curriculum of CS Study Program (Curriculum 2021), Faculty of Engineering Technology (FET), SU.
4. Input and suggestions from teaching staff in CS Study Program, FET, SU.
5. Input and suggestions from advisory board: Industrial Advisory Board, academics, and alumni.
6. Benchmarking with local and international universities.
7. Panduan Penyusunan Kurikulum Pendidikan Tinggi Mendukung Merdeka Belajar-Kampus Merdeka Menuju Indonesia Emas, Direktorat Jenderal Pendidikan Tinggi Riset dan Teknologi, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi, 2024.
8. Buku Panduan Merdeka Belajar-Kampus Merdeka 2024, Direktorat Jenderal Pendidikan Tinggi Riset dan Teknologi, Kementerian Pendidikan, Kebudayaan, Riset, dan Teknologi, 2024.

Based on the above consideration, the 2024 Curriculum of CS Study Program is designed to produce graduates that can compete globally. The CS program graduates are emphasized to possess logical, critical, algorithmic thinking, analysis skill, able to design, implement, and assess computing solutions, adhere to professional ethics, and have the initiative to conduct lifelong learning to continuously innovate and create useful technologies for society.

2. Vision, Mission, Purpose, and Strategy of Study Program

2.1. Vision

University's Vision	Faculty's Vision
Sampoerna University aspires to foster future leaders with strong moral character and internationally competitive skills set, enabling them to actively participate in building a more prosperous, equitable, respected, and globally competitive Indonesia.	The Faculty of Engineering and Technology at Sampoerna University aspires to become a Faculty that is internationally competitive in the field of engineering and technology by 2030 whose graduates are highly sought after by industries, are entrepreneur minded, are socially responsible, and are competent to pursue graduate degrees.

CS Study Program's Vision:

To become an established Computer Science study program whose graduates are equipped with competitive skills sets in the field of computing, strong moral character, are entrepreneur minded, are socially responsible in building a more prosperous and globally competitive Indonesia by 2030.

CS Study Program's Scientific Vision:

To become a Computer Science study program that is advancing the field of software engineering, artificial intelligence, and providing integrated and multi-platforms computing solutions.

2.2. Mission

University's Mission	Faculty's Mission
Sampoerna University aims to provide students with affordable access to education that meets the highest international standards. Sampoerna University offers a curriculum unique in Indonesia built around an American general education core, successfully preparing students for credential completion at Sampoerna University or for recognized transfer abroad. We also provide a full spectrum of co-curricular and pre-professional opportunities that ensure student success, preparing leaders for a global society.	The Faculty of Engineering and Technology at Sampoerna University strives To provide an education in engineering and technology having international standards, built around an American general education core, that prepares its graduates as highly competent, entrepreneur minded, and socially responsible leaders; and are committed to giving back to Indonesia and society.

- To conduct innovative research in the field of engineering and technology for the benefits of humankind.

CS Study Program's Mission:

- Improving the quality of higher education in the field of Computer Science based on research and innovation in order to increase the nation's competitiveness;
- Improving the quality of research and innovation as well as conducting publications as an effort to develop Computer Science by prioritizing values, ethics, norms, and culture;
- Improving the quality of community service as an effort to apply and develop computer science to society based on research and innovation;
- Improving the quality of governance in managing the Computer Science Study Program.

2.3. Purpose

University's Purpose	Study Program's Purpose
Sampoerna University aims to provide students with affordable access to education that meets the highest international standards.	1. To produce qualified and highly competitive Computer Science graduates to become entrepreneurs and leaders with ethics and integrity. 2. To produce Computer Science graduates who have international insight and are able to communicate and compete in the international job market. 3. To produce research and scientific works that can contribute to the development of computer science and technology, the needs of industry and business, as well as the academic world at the national, regional, and international levels. 4. To produce Computer Science graduates who have a sense of and responsibility for national economic development to improve the quality of life through community service programs.

2.4. Strategy

University's Strategy	Study Program's Strategy
Sampoerna University employs four pillars: 1. Reimagine and maintain our core curriculum. This is done to identify skills and competencies of graduates and to review and strengthen Sampoerna University's core curriculum. 2. Create an Institution of and for our students. This is done by recruiting and preparing qualified students to enroll and succeed in Sampoerna University, also to engage students in university governance, policy development, and external relations. 3. Build toward the future of Indonesia. This is done to develop students' capacity to contribute to the nation. 4. Provide a model of tomorrow's university today. This is to ensure affordability and access to education that increases in value.	To ensure graduates of CS Study Program upholding Sampoerna University's values (see section 2.5), the graduate profiles of CS Study Program are first derived by considering university's values. These graduate profiles are then derived into PLO (Program Learning Outcomes), Body of Knowledge, Courses, and Subject Learning Outcome. In focusing on the American style general education, CS Study Program adopts General Education Sampoerna University Core Curriculum. Meanwhile, CS Study Program also adopts the guidance from APTIKOM in developing its curriculum. In ensuring the achievement of the graduate profiles, CS Study Program measures the learning outcome for each course and develops assessments to measure the achievement of the learning outcomes.

2.5. University Value

Sampoerna University strives to produce graduates with a strong moral character and has sufficient skill sets to contribute to the nation. Sampoerna University values the contribution or giving to the nation as the highest goal in all aspects. Historically, Putera Sampoerna Foundation, which owns Sampoerna University, has had a strong responsibility in contributing to the nation and this is shown greatly in the vision and mission of the university. In supporting this value, CS Study Program strives to equip the graduates with competitive skill sets and strong moral character to be able to contribute to the nation.

In general, following are the values that are instilled to the graduates of Sampoerna University:

1. Integrity → Accountability, intellectual integrity, and respect for human values and social ethics.

2. Growth → Intellectually curious to expand their competencies and courageously take initiative, sharing expertise as responsible leaders in the community.
3. Knowledge → Authoritative and internationally articulate in their field, while continuing to be at the forefront of expanding knowledge, especially in modern science and technology.
4. Inspiration → Able to inspire individuals and communities through effective communication, critical thinking and problem solving.
5. Tenacity → Determined and persistent in achieving goals.
6. Empowerment → Has entrepreneurial spirit and be creative in achieving outcomes.

These values are taught by focusing on an American-style general education which is a distinctive value of academic programs at Sampoerna University. By referring to these values and the distinctive American-style general education, CS Study Program then developed the 2024 Curriculum.

3. Curriculum Evaluation & Tracer Study

3.1. Curriculum Evaluation

The curriculum was evaluated to include the needs of graduate users such as the industries that have not been covered in the previous curriculum, learning process, research, and the society service at the faculty level. Usually, the change in the curriculum is expected to take place once every 5 years. However, the curriculum review of CS Study Program can be done before 5 years if there are fundamental changes related to the graduate competencies.

The 2024 CS Curriculum is the result of the 2021 CS Curriculum evaluation. The review of the 2021 Curriculum was done because of the requirement to obtain international accreditation that employs outcome-based education (OBE) curriculum. The followings are the steps in evaluating the curriculum in CS Study Program of Sampoerna University:

1. The Dean of Faculty of Engineering and Technology issues a Dean's decree to form the Curriculum Development team in each study program.
2. The Head of Study Program holds evaluation meetings on the ongoing curriculum with all the teaching staff in the study program.
3. The Head of Study Program forms the schedule and the curriculum development team.
4. The Head of Study Program with the curriculum development team develop the curriculum framework which refers to: (i) discussion meeting with teaching staffs, (ii) recent trends in science, technology, and information, (iii) government's law, (iv) vision and mission of the university, faculty, and study program, (v) graduates' user, (vi) stakeholders' opinion (Academia, Industrial Advisor Board, Students, Alumni), and (vii) benchmarking with related universities.
5. The Head of Study Program holds meetings on the curriculum concepts by inviting all stakeholders, i.e., permanent lecturers, non-permanent lecturers, practitioners, and graduates' users to obtain input.

6. The Head of Study Program improves the curriculum based on input from several stakeholders and prepares the presentation.
7. The curriculum is presented at the faculty and university level to obtain the approval from the Vice Rector of Academic Affairs.
8. The Head of Study Program finalizes the concept of the curriculum to be the document that is ready to be validated.
9. The Dean of FET submitted a letter to the Vice Rector of Academic Affairs to validate the curriculum.
10. The Rector signs the validation of the curriculum.
11. Once the curriculum is validated, the curriculum is socialized to all lecturers and students.
12. The new curriculum is implemented.

In evaluating the curriculum of the CS Study Program, the flowchart shown in Figure 1 is used as a reference. In the flowchart, the curriculum development starts from the definition of the Graduate Profile (GP). The GP is formulated based on the market needs considering the university's vision. In detail, following are the mechanism in evaluating the 2021 Curriculum at the study program:

1. The market needs of computer science graduates are evaluated by getting the inputs from the IAB (Industrial Advisory Board) of CS study program. These inputs are then made as a reference to evaluate the GP of the study program. The GP is also evaluated and developed by considering the university's vision. In addition, the input from the alumni and the students are also considered. Meanwhile, surveys were carried out to the current students to obtain their input on the strengths and weaknesses of the current courses.
2. After the GP is evaluated, the PLOs (Program Learning Outcomes) are evaluated. The PLOs are defined as the outcomes necessary to be achieved so that graduates have the characteristics of the GP. Benchmarking with other universities was also performed to evaluate the PLOs. The PLOs are finalized with curriculum team meetings with the Head of Study Program.
3. After PLOs' evaluation, mapping of PLOs to the courses is done. This evaluation is done by technical meeting of teaching staff in the study program by considering rules from the university pertaining courses.
4. The next step is to evaluate the SLO (Subject/Student Learning Outcome) which is developed using Bloom's Taxonomy. This is the outcome setting for each course which must refer to the PLOs related to the course. To bridge the PLO and SLO, we define Signature Assignment as performance indicator for each course.
5. The SLO is one of the contents in the Syllabus / lesson plan / RPS (*Rencana Pembelajaran Semester*). The RPS explains about the outcome expected from the course and the learning plan in one semester. The RPS for all courses are then evaluated thoroughly to conform to the newly defined PLOs and SLOs.

6. The last step is to develop the operational guidance for MBKM (*Merdeka Belajar Kampus Merdeka*).



Figure 1. Flowchart of the curriculum development

Following is the summary of changes/evaluations done to the curriculum of CS Study Program of SU:

1. There are 10 PLOs in the 2024 Curriculum adopting the PLOs recommended by APTIKOM, compared to 8 PLOs in the previous curriculum.
2. Credit reductions in some of the General Education courses, i.e., Mathematics and Physics courses, of which each course previously had 5 credits. In the new curriculum, these courses are now only 3 – 4 credits each. This allows the addition of more CS courses into the curriculum.

3. The existence of some digital electronics / circuits related courses will serve as the foundations toward IoT (Internet of Things), which combines hardware, software, and internet, to answer the needs of Industrial Revolution 4.0, in which IoT plays an important part.
4. The RPS are reviewed and redeveloped based on the defined SLOs, referring to new PLO. These change some of the assessments, including the signature assignment, in the course.
5. The implementation of MBKM is included in the 2024 Curriculum as well.

3.2. Tracer Study

CS Study Program is currently preparing the necessary instrument for the graduates' tracer study which is integrated with the university handled by SAA (Student and Alumni Affairs) of Sampoerna University. There are two types of questionnaires. The first type of questionnaire is distributed to the Alumni to obtain feedback for the study program whether the Alumni thinks that they have achieved the Program Learning Outcomes (PLOs). The second type of questionnaire is distributed to the graduates' user (companies where our graduates work), essentially to obtain feedback whether the Alumni have achieved the specified PLOs. The second type of questionnaire is also used to gather information from the graduates' user about the skills needed by the graduates in the working field.

CS Study Program strives to improve the performance and quality of the study program to produce graduates with skills that are needed by industry and society and able to compete globally. CS Study Program is also directing the graduates to open their own startup company to open employment opportunities for society.

4. Graduate Profile (GP) & Program Learning Outcomes (PLOs)

4.1. Graduate Profile

The Graduate Profile (GP) or PEO (Program Educational Objectives) of the CS Study Program is developed based on the inputs from the stakeholders, as described in **Table 1**.

Table 1. Graduate Profile (GP) of CS Study Program

GP Code	Graduate Profile
GP - 1	Graduates are capable to think logically, critically and systematically in utilizing computer science/informatics knowledge to solve real world problems.
GP - 2	Graduates possess the skills to analyze computing issues and utilize computing principles along with other relevant fields to find solutions for organizations.

GP - 3	Graduates are capable of designing, implementing, and assessing computing solutions that appropriately meet user needs / requirements.
GP - 4	Graduates adhere to legal standards, socio-cultural norms, and professional ethics.
GP - 5	Graduates are dedicated to ongoing self-improvement through lifelong learning.

Our Graduate Profile reflects the expected career outcomes for Computer Science graduates as listed in **Table 2**.

Table 2. Expected Profession based on GP

No	Professional Field	Job Role
1	Programming, Software Development and Software Engineer	Software Developer/ Engineer, Program Analyst, Web Developer, DevOps Engineer, Full Stack Developer, Quality Assurance (QA) Engineer, Mobile App Developer, etc.
2	Artificial Intelligence (AI) Engineer	Data Scientist, Machine Learning Engineer, AI Research Scientist, AI Engineer, AI Software Engineer, AI Ethics Specialist, Natural Language Processing (NLP) Engineer, Computer Vision Engineer, etc.
3	Digital Peripherals and IoT Engineer	IoT Developer, Robotics Engineer, Embedded Systems Engineer, IoT System Architect, Hardware Engineer, Circuit Design Engineer, Network Security Analyst, Digital Computer Technology Advisor, etc.
4	Instructor and Research Assistant in Computer Science / Informatics	Teaching Assistant, Lab Assistant, Lab Technician, Research Assistant, Research Fellow, etc.

4.2. Formulation of PLOs (Program Learning Outcomes)

PLOs of the Study Program are defined to ensure the graduates have the characteristics mentioned in the GP (see **Table 3**). The PLOs are defined based on the guidance from APTIKOM. The PLOs are also defined by considering the outcomes based on KKNi's criteria.

CS Study Program, like other study programs at Sampoerna University, adopts the General Education Sampoerna University Core (Gen-Ed SU Core) Curriculum for the basic science courses and general courses. These courses are mathematics, science (physics, chemistry) and general courses such as Bahasa Indonesia, English language courses, Religion, etc. To ensure the achievement of the PLOs of Study Program, a mapping matrix between the PLOs of Gen-Ed SU Core and PLOs of Study Program is needed.

The relation between PLOs of the 2024 Curriculum and PLOs from IABEE, KKN1, and Gen-Ed SU Core can be seen in **Tables 4 through 6**.

Table 3. PLOs of CS Study Program

No	Description of PLO	GP1	GP2	GP3	GP4	GP5
PLO 1	Mastering competencies in algorithms and applied mathematics to innovate and solve complex problems.	✓	✓			
PLO 2	Possess a comprehensive understanding of computer system operations and the ability to apply various algorithms and methods / techniques to solve organizational problems.	✓	✓			
PLO 3	Exhibit a strong understanding of theoretical and practical concepts in Computer Science/Informatics to design, simulate, and implement multi-platform technology applications that address the evolving needs of industry and society.	✓		✓		
PLO 4	Capable of designing, implementing and evaluating computing-based solutions that meet computing needs by considering the most effective methods and algorithms.		✓	✓		
PLO 5	Proficient in designing, implementing, and evaluating multi-platform computing solutions aligned with organizational computing requirements.		✓	✓		
PLO 6	Able to design, develop, and optimize algorithms for various advanced applications, particularly in the field of Programming and Software Development, Artificial Intelligence (AI), Digital Peripherals and IoT.	✓	✓	✓		
PLO 7	Demonstrate competence in analyzing complex computing problems to identify and implement effective IT project management solutions, integrating insights from developing of transdisciplinary knowledge.	✓	✓		✓	✓
PLO 8	Able to analyze, design, create, and evaluate interactive applications and the user interface by prioritizing user needs, optimizing user experience, and leveraging transdisciplinary knowledge.			✓	✓	✓

PLO 9	Demonstrate professional responsibility and the ability to make informed decisions through lifelong learning in computing practices, adhering to legal, ethical, and societal standards.					✓	✓
PLO 10	Possess leadership, teamwork, and communication skills, with the ability to effectively present and manage tasks in dynamic and innovative environments.					✓	✓

Table 4. Relation between PLOs of CS Study Program and PLOs recommended by IABEE

No	PLOs from IABEE	PLO									
		1	2	3	4	5	6	7	8	9	10
1	Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	✓	✓		✓		✓	✓	✓		
2	Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.			✓	✓	✓	✓	✓	✓		
3	Communicate effectively in a variety of professional contexts.									✓	✓
4	Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.									✓	✓
5	Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.									✓	✓
6	Apply computer science theory and software development fundamentals to produce computing-based solutions.	✓	✓	✓		✓	✓	✓	✓		

Table 5 shows the mapping of CS Study Program PLOs with those of KKNI.

Table 5. Relation between PLOs of CS Study Program and PLOs based on KKNI

	<i>Sikap / Attitude</i>	PLO									
		1	2	3	4	5	6	7	8	9	10
S-1	Devoted to God Almighty and able to present a religious attitude;									✓	
S-2	Uphold humanistic values in carrying out duties based on religion, morals, and ethics;									✓	
S-3	Contribute in improving the quality of the community, citizenship and civilized progress based on the Pancasila;									✓	✓

S-4	Act as a citizen who is proud and loves his/her homeland, nationalistic and has a sense of responsibility to the state and the nation;									✓	
S-5	Appreciate cultural diversity, views, religions, beliefs, and other people's original opinions;									✓	
S-6	Able to cooperate, have social sensitivity, and care about the community and the environment;										✓
S-7	Obey the law and discipline in being part of the community and in exercising citizenship;									✓	
S-8	Internalize values, norms, and academic ethics;									✓	
S-9	Demonstrate responsible attitude towards work in the field of expertise independently; and									✓	
S-10	Internalize the spirit of independence, tenacity, and entrepreneurship.										✓
		PLO									
	Penguasaan Pengetahuan / Knowledge	1	2	3	4	5	6	7	8	9	10
PP-1	Mastering the theoretical concepts of computer science/informatics in general and the theoretical concepts of specific parts in the field of knowledge in depth and formulating problem-solving;	✓		✓							
PP-2	Demonstrate comprehensive knowledge of computer systems with the ability to design and develop algorithms or methods for problem-solving;		✓		✓		✓				
PP-3	Demonstrate knowledge in developing algorithms or methods for implementation in computer-based software;		✓		✓		✓				
PP-4	Ability to analyze complex computing problems, apply computing principles and relevant disciplines, and incorporate insights from transdisciplinary developments to identify solutions; and			✓	✓	✓		✓			
PP-5	Mastering algorithms and mathematics competencies.	✓		✓	✓						
	Keterampilan Umum / General Skills	PLO									

		1	2	3	4	5	6	7	8	9	10
KU-1	Able to implement logical, critical, systematic, and innovative thinking, within the context of development or implementation of science and technology that cares and applies the appropriate humanistic values related to his/her area of expertise;	✓						✓	✓		
KU-2	Able to demonstrate the ability to work independently, with quality and measurable performance;									✓	
KU-3	Able to review developmental areas or science and technology implementations which applies humanistic values in accordance with his/her expertise based on scientific rules, procedures, and ethics in order to improve solutions, ideas, designs or artistic criticisms;			✓		✓					
KU-4	Capable of preparing scientific descriptions of the results produced in a thesis form or final project report, and upload them into the higher education website;										✓
KU-5	Capable of making decisions appropriately to solve problem in his/her area of expertise based on the analysis of information and data;							✓	✓	✓	
KU-6	Able to maintain and develop his/her working network with advisors, colleagues, and peers in both inside and outside his/her institution;										✓
KU-7	Capable of taking responsibility for the group's work achievement by conducting appropriate supervision and evaluation towards work completion assigned to those who work under his/her responsibility;										✓
KU-8	Able to conduct self-evaluation process related to the work of the group who is under his/her responsibility, and manage independent learning;									✓	✓
KU-9	Capable of documenting, storing, securing, and rediscovering data to ensure validity and prevent plagiarism;						✓				
KU-10	Able to define user or market needs for performance by analyzing, evaluating, and developing computer-based algorithms or methods;			✓	✓	✓	✓		✓		
KU-11	Demonstrate managerial teamwork skills, self-management, effective									✓	✓

	communication (verbal and written), and strong presentation abilities; and										
KU-12	Demonstrate life skills at undergraduate (<i>Sarjana</i>) program level.									✓	✓
	Keterampilan Khusus / Specific Skills	PLO									
		1	2	3	4	5	6	7	8	9	10
KK-1	Able to design and develop algorithms for various needs;		✓		✓		✓				
KK-2	Able to analyze, design, create and evaluate user interfaces and interactive applications by considering user needs and experience;			✓	✓	✓			✓		
KK-3	Able to design, develop, and simulate multi-platform technology applications relevant to industry needs; and			✓		✓					
KK-4	Able to design, implement and evaluate computing-based solutions that meet the computing needs of a program discipline.				✓	✓		✓			

Table 6 maps the CS PLOs to the PLOs of General Education (Gen-Ed) courses provided at Sampoerna University. **Table 7** provides the mapping of SU courses to the associated PLOs.

Table 6. Relation between PLOs of Study Program and PLOs of Sampoerna University's General Education (Gen-Ed)

No	PLOs from Gen-Ed SU Core										
		1	2	3	4	5	6	7	8	9	10
1	Critical Thinking (CT)	✓			✓	✓		✓			
2	Effective Communication (EC)										✓
3	Ethical Reasoning (ER)									✓	
4	Global Self Awareness (GSA)									✓	✓
5	Information Literacy (IL)		✓			✓	✓		✓	✓	
6	Mathematical & Scientific Reasoning (MSR)	✓		✓	✓	✓	✓				

Table 7. The Mapping of Courses to the Associated PLOs

Semester	Courses			PLO									
	Course Code	Course Name	Credits	1	2	3	4	5	6	7	8	9	10
1	GCOM1304	Composition I	3										✓
	GENG1301	Introduction to Engineering	3	✓						✓			
	GSCI1303	General Chemistry I	3	✓								✓	
	GSCI1103L	General Chemistry I Lab	1	✓									
	GMAT1313	Calculus I	3									✓	
	GSOC2304	Social Problems	3									✓	
	GHUM1207	Kewarganegaraan	2										✓
	TOTAL		18										
2	GCOM1305	Composition II	3										✓
	GMAT1314	Calculus II	3	✓									
	COMP1401	Computer Programming	4	✓					✓				
	GSCI1312	General Physics with Calculus I	3	✓									
	GSCI1112L	General Physics with Calculus I Lab	1									✓	
	GHUM1303	World Religions	3									✓	
	GHUM1301	Pancasila	3										✓
	TOTAL		20										
3	GMAT2415	Calculus III	4	✓									
	GSCI2313	General Physics with Calculus II	3	✓									
	GSCI2113L	General Physics with Calculus II Lab	1	✓	✓				✓				
	COMP2424	Data Structures & Algorithms	4		✓				✓				
	COMP2303	Digital Systems	3	✓									
	MTED3314	Discrete Mathematics	3										✓
	GHUM1208	Indonesian Language	2										✓
	TOTAL		20										
4	COMP1402	Object Oriented Programming	4	✓					✓				
	GSCI2210	Introduction to Optics and Thermodynamics	2	✓									
	COMP2304	Computer Organization & Architecture (incl. Introduction to OS)	3		✓		✓						
	GMAT2308	Intro. to Ordinary Differential Equations	3	✓									
	VCDD1326	Design Thinking & Theory	3								✓		
	GSOC 2303	General Psychology	3									✓	
	TOTAL		18										
5	COMP3524	Basic Circuits (with lab)	5						✓				
	ENGR3303	Environmental Ethics	3									✓	
	COMP3408	Database Management Systems (incl. data visual.)	4					✓					
	COMP3423	Mathematics for Engineering Applications	4	✓									

	COMP3311	Software Engineering	3		✓														
		TOTAL	19																
6	COMP3326	Circuit Theory	3						✓										
	COMP4429	Microprocessor Organization (with Lab)	4	✓															
	COMP3309	Computer Networks	3							✓									
	GSOC1301	Prin. of Macroeconomics	3												✓				
	COMP3312	Human-Computer Interaction (incl. UI/UX Design)	3										✓						
	COMP3315	Artificial Intelligence (incl. Intro to Machine Learning)	3						✓										
		TOTAL	19																
7	COMP3325	Computer Architecture & Design	3	✓															
	COMP4427	Electronic Circuits (with lab)	4						✓										
	COMP4328	Intro. To Communications	3							✓									
	COMP4322	Mobile Apps Development	3		✓														
	COMP4333	Internship for Computer Science	3														✓		
	COMP4334	Final Year Project 1 for CS	3					✓										✓	
		TOTAL	19																
8	COMP4330	Knowledge System Engineering	3		✓														
	COMP4331	Fundamentals of Info. & Network Security	3													✓			
	COMP4332	Computer Aided Logic Design	3						✓										
		Elective:																	
	COMP2306	Operating Systems	3			✓													
	COMP3313	Web Programming					✓												
	INFS2318	Technopreneurship								✓									
	COMP4335	Final Year Project 2 for CS	3					✓										✓	
		TOTAL	15																
		TOTAL MINIMUM CREDITS	148																

5. Body of Knowledge (BoK)

This section explains the mapping of CPL and relevant Body of Knowledge. Body of knowledge can be field of study, or parts of it. Each of body of knowledge should be given a code to map them with CPL.

In general, Computer Science BoK are derived from Bahan Kajian in APTIKOM's OBE KKNi Curriculum Development Guideline 2024 which are based on ACM/IEEE CC 2020, CS 2023, SN DIKTI, ASIIN, and IABEE. **Table 8** shows the mapping of the CS BoK to the PLOs, **Table 9** shows the list of CS BoK and the associated courses, and **Table 10** shows the mapping of courses to the associated BoK.

Table 8. Relation between CS BoK and PLO of CS Study Program

No	Computer Science (CS) BoK	PLO of CS Study Program									
		1	2	3	4	5	6	7	8	9	10
BoK1	Society, Ethics and Professionalism									✓	✓
BoK2	Security		✓				✓			✓	
BoK3	Project Management				✓			✓			✓
BoK4	User Experience Design				✓	✓			✓		
BoK5	Software Development Fundamentals		✓	✓		✓	✓				
BoK6	Data Management		✓	✓	✓						
BoK7	Parallel and Distributed Computing		✓				✓				
BoK8	Network and Communication		✓				✓				
BoK9	Human-Computer Interaction				✓		✓		✓		
BoK10	Software Engineering	✓	✓		✓	✓		✓			
BoK11	Operating Systems	✓	✓		✓	✓	✓				
BoK12	Algorithmic Foundations	✓	✓				✓				
BoK13	Foundations of Programming Languages	✓	✓		✓	✓	✓				
BoK14	Programming Fundamentals	✓	✓		✓	✓	✓				
BoK15	Systems Fundamentals		✓		✓	✓	✓				
BoK16	Architecture and Organization		✓		✓	✓					
BoK17	Graphics and Interactive Techniques			✓	✓		✓		✓		
BoK18	Artificial Intelligence	✓	✓	✓		✓	✓	✓			
BoK19	Specialized Platform Development		✓	✓	✓	✓					
BoK20	Mathematical and Statistical Foundations	✓	✓	✓							
BoK21	Character Building									✓	✓
BoK22	Methodology	✓	✓		✓			✓			

Table 9. Computer Science (CS) Body of Knowledge

	Body of Knowledge (BoK)	BoK Code	List of Associated Courses
1	Society, Ethics and Professionalism	BoK-1	Social Problems General Psychology Indonesian Language Environmental Ethics

	Body of Knowledge (BoK)	BoK Code	List of Associated Courses
			Technopreneurship Composition I Composition II Principle of Macroeconomy General Chemistry I General Chemistry I Lab
2	Security	BoK-2	Fundamentals of Info. & Network Security
3	Project Management	BoK-3	Internship for Computer Science Final Year Project I Final Year Project II
4	User Experience Design	BoK-4	Design Thinking & Theory (GE EP-Artist) Human-Computer Interaction (incl. UI/UX Design)
5	Software Development Fundamentals	BoK-5	Web Programming
6	Data Management	BoK-6	Database Management Systems (incl. data visual.) Data Structures & Algorithms
7	Parallel and Distributed Computing	BoK-7	Computer Networks
8	Network and Communication	BoK-8	Intro. To Communications Computer Networks
9	Human-Computer Interaction	BoK-9	Human-Computer Interaction (incl. UI/UX Design)
10	Software Engineering	BoK-10	Software Engineering
11	Operating Systems	BoK-11	Operating Systems Computer Organization & Architecture (incl. Intro to OS)
12	Algorithmic Foundations	BoK-12	Data Structures & Algorithms
13	Foundations of Programming Languages	BoK-13	Object Oriented Programming Computer Programming Software Engineering Mobile Apps Development Web Programming
14	Programming Fundamentals	BoK-14	Computer Programming Object Oriented Programming
15	Systems Fundamentals	BoK-15	Intro to Engineering Computer Organization & Architecture (incl. Intro to OS) Digital Systems
16	Architecture and Organization	BoK-16	Computer Organization & Architecture (incl. Intro to OS) Operating System Computer Architecture & Design

	Body of Knowledge (BoK)	BoK Code	List of Associated Courses
			Computer Aided Logic Design Digital Systems Basic Circuits Circuit Theory Electronic Circuits (with lab) Microprocessor Organization (with Lab)
17	Graphics and Interactive Techniques	BoK-17	Design Thinking & Theory (GE EP-Artist) Human-Computer Interaction (incl. UI/UX Design)
18	Artificial Intelligence	BoK-18	Knowledge System Engineering Artificial Intelligence
19	Specialized Platform Development	BoK-19	Mobile Apps Development
20	Mathematical and Statistical Foundations	BoK-20	Calculus I Calculus II Calculus III Discrete Mathematics Intro. to Ordinary Differential Equations Mathematics for Engineering Applications General Physics with Calculus I General Physics with Calculus II Introduction to Optics & Thermodynamics
21	Character Building	BoK-21	Pancasila World Religions Environmental Ethics Kewarganegaraan
22	Methodology	BoK-22	Final Year Project I Final Year Project II

Table 10. The Mapping of Courses to the Associated BoK

Semester	Courses			Body of Knowledge (BoK)																			
	Course Code	Course Name	Credits	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	GCOM1304	Composition I	3	✓																			
	GENG1301	Introduction to Engineering	3															✓					
	GSCI1303	General Chemistry I	3	✓																			
	GSCI1103L	General Chemistry I Lab	1	✓																			
	GMAT1313	Calculus I	3																			✓	
	GSOC2304	Social Problems	3	✓																			
	GHUM1207	Kewarganegaraan	2	✓																			✓
TOTAL			18																				
2	GCOM1305	Composition II	3		✓																		
	GMAT1314	Calculus II	3																			✓	
	COMP1401	Computer Programming	4													✓	✓						
	GSCI1312	General Physics with Calculus I	3																			✓	
	GSCI1112L	General Physics with Calculus I Lab	1																			✓	
	GHUM1303	World Religions	3																				✓
	GHUM1301	Pancasila	3																				✓
TOTAL			20																				
3	GMAT2415	Calculus III	4																			✓	
	GSCI2313	General Physics with Calculus II	3																			✓	
	GSCI2113L	General Physics with Calculus II Lab	1																			✓	
	COMP2424	Data Structures & Algorithms	4					✓							✓								
	COMP2303	Digital Systems	3															✓	✓				
	MTED3314	Discrete Mathematics	3																			✓	
	GHUM1208	Indonesian Language	2	✓																			
TOTAL			20																				

KPT 4.0 - 30

	COMP4334	Final Year Project 1 for CS	3		✓														✓
		TOTAL	19																
8	COMP4330	Knowledge System Engineering	3														✓		
	COMP4331	Fundamentals of Info. & Network Security	3	✓															
	COMP4332	Computer Aided Logic Design	3													✓			
	COMP2306	Elective: Operating Systems	3							✓						✓			
	COMP3313	Web Programming					✓					✓							
	INFS2318	Technopreneurship		✓															
	COMP4335	Final Year Project 2 for CS	3		✓														✓
		TOTAL	15																
		TOTAL MINIMUM CREDITS	148																

6. Distribution of Courses in Each Semester

Table 11 below shows the course distribution in each semester, and the flow of courses is shown in **Figure 2**. The elective courses may vary in the future, including elective courses from other study programs.

Table 11. CS Study Program Course Distribution

SEMESTER I

CODES	COURSE TITLE	CREDITS
GCOM1304	Composition I	4
GENG1301	Introduction to Engineering	3
GSCI1303	General Chemistry I	3
GSCI1103L	General Chemistry I Lab	3
GMAT1313	Calculus I	5
GSOC2304	Social Problems	3
GHUM1207	Kewarganegaraan	2
Total		18

SEMESTER II

CODES	COURSE TITLE	CREDITS
GCOM1305	Composition II	1
GMAT1314	Calculus II	3
COMP1401	Computer Programming	4
GSCI1312	General Physics with Calculus I	3
GSCI1112L	General Physics with Calculus I Lab	1
GHUM1303	World Religions	3
GHUM1301	Pancasila	3
Total		20

SEMESTER III

CODES	COURSE TITLE	CREDITS
GMAT2415	Calculus III	4
GSCI2313	General Physics with Calculus II	3
GSCI2113L	General Physics with Calculus II Lab	1
COMP2424	Data Structures & Algorithms	4
COMP2303	Digital Systems	3
MTED3314	Discrete Mathematics	3
GHUM1208	Indonesian Language	2
Total		20

SEMESTER IV

CODES	COURSE TITLE	CREDITS
COMP1402	Object Oriented Programming	4
GSCI2210	Introduction to Optics and Thermodynamics	2
COMP2304	Computer Organization & Architecture (incl. Intro to OS)	3
GMAT2308	Intro. to Ordinary Differential Equations	3
VCDD1326	Design Thinking & Theory	3
GSOC 2303	General Psychology	3
Total		18

SEMESTER V

CODES	COURSE TITLE	CREDITS
COMP3524	Basic Circuits (with lab)	5
ENGR3303	Environmental Ethics	3
COMP3408	Database Management Systems (incl. data visualization)	4
COMP3423	Mathematics for Engineering Applications	4
COMP3311	Software Engineering	3
Total		19

SEMESTER VI

CODES	COURSE TITLE	CREDITS
COMP3326	Circuit Theory	3
COMP4429	Microprocessor Organization (with Lab)	4
COMP3309	Computer Networks	3
GSOC1301	Principle of Macroeconomics	3
COMP3312	Human-Computer Interaction (incl. UI/UX Design)	3
COMP3315	Artificial Intelligence (incl. Intro to Machine Learning)	3
Total		19

SEMESTER VII

CODES	COURSE TITLE	CREDITS
COMP3325	Computer Architecture & Design	3
COMP4427	Electronic Circuits (with lab)	4
COMP4328	Intro. To Communications	3
COMP4322	Mobile Apps Development	3
COMP4333	Internship for Computer Science (activity in Summer)	3
COMP4334	Final Year Project 1 for CS	3
Total		19

SEMESTER VIII

CODES	COURSE TITLE	CREDITS
COMP4330	Knowledge System Engineering	3
COMP4331	Fundamentals of Info. & Network Security	3
COMP4332	Computer Aided Logic Design	3
COMP2306 COMP3313 INFS2318	Elective: Operating Systems Web Programming Technopreneurship	3
COMP4335	Final Year Project 2 for CS	3
Total		15
TOTAL CREDIT HOURS FOR CS DEGREE PROGRAM		148

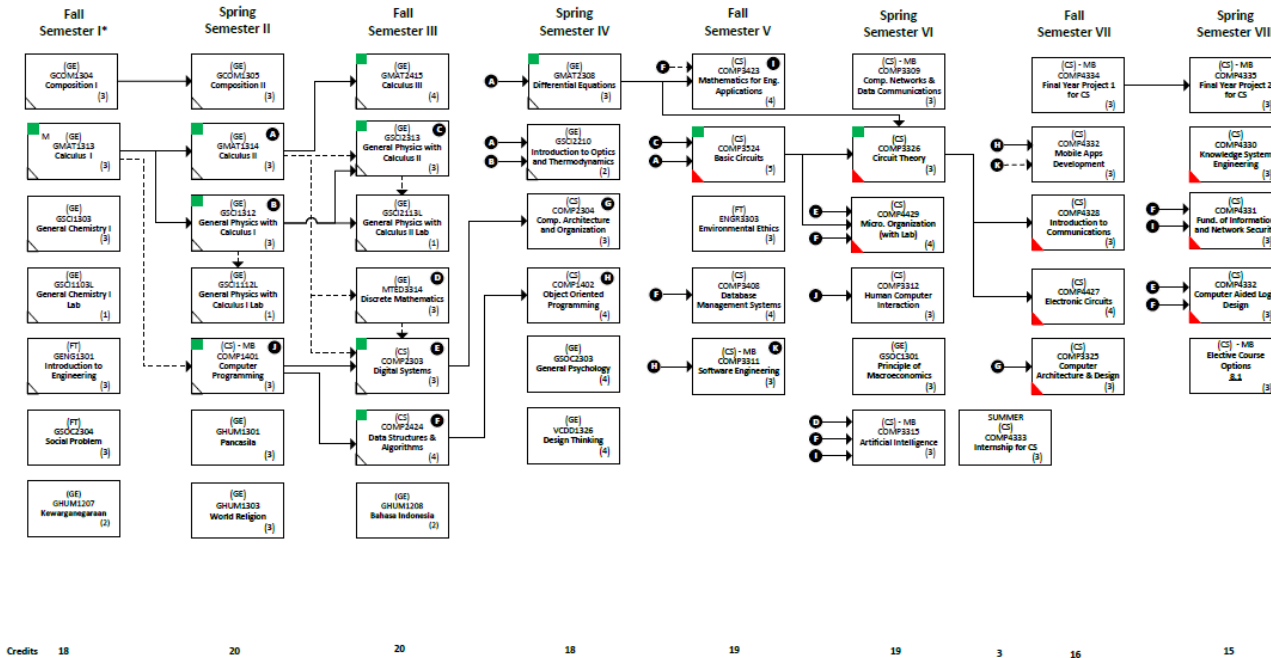
CURRICULUM DOCUMENT

COMPUTER SCIENCE STUDY PROGRAM

TOTAL CREDIT HOURS : 148

This flowchart is based on 2024/2025 curriculum and remains in effect until the next flowchart is published

COMPUTER SCIENCE



Flowchart represent a recommended eight semester path to graduation of your respective program and respective degree.

ELECTIVE COURSE OPTIONS

- B.1**
a. COMP2306 Operating Systems (3) - MB
b. COMP3313 Web Programming (3)
Pre-requisite for point b : COMP1401 Computer Programming (3)
Co-requisite for point b : COMP3408 Database Management Systems (4)
c. INF52316 Technopreneurship

FLOWCHART LEGEND

- * College Readiness Requirement Applies (See College Readiness Table)
M Math Placement is Required (See College Readiness Table)
----> Pre-requisite
-----> Co-requisite
General Education courses are co-develop in collaboration with Broward College
- Green box: Critical Path to 8 Semester Graduation
Red box: Critical Courses That Must Be Completed in Upper Division
Blue box: Critical Courses That Must Be Completed in Lower Division
MB Non-Majors May Earn MBKM Credits for This Course
(GE) General Education Courses
(FT) Faculty of Engineering & Technology Courses
(CS) Computer Science Courses

FLOWCHART APPROVAL

Approved by

HOD Dean

Disclaimer: courses and flowchart can change based on requirements of each degree

Figure 2. Flow of Courses in Computer Science Study Program Sampoerna University

7. MBKM Implementation Plan

Referring to the *PermenDikBud No. 3 Tahun 2020 (Standar Nasional Pendidikan Tinggi)*, *Pasal 15 and 18 about MBKM (Merdeka Belajar dan Kampus Merdeka)*, the MBKM credits recognition can be implemented by:

A. Maximum of 20 credits are taken at other study programs in SU, and

B. Maximum of 40 credits are taken at:

- Computer Science Study Program in different universities, and/or
- Different study program in different universities, and/or
- Internship at company or recognized institution.

Considering the 2024 CS curriculum (summarized in **Table 11**) and referring to the SU Rector's decree on the implementation of MBKM in Sampoerna University, as well as the policy of General Education at Sampoerna University, the following sections explain the implementation model of MBKM in CS study program.

A. Maximum of 20 credits could be taken at other study programs in SU, they are:

- **6 credits could be fulfilled by** taking courses at other study programs in SU, they are:
 - Mathematics course
Students can take GMAT2308 – Differential Equations course (3 credits) at other study programs in SU.
 - Art course
Students can choose **one** out of the following courses: Basic Fundamentals of Drawings (3 credits), Wayang Studies (3 credits), Indonesia Art History (3 credits), and Design Thinking & Theory (3 credits) from Visual Communication Design study program.
- **14 credits could be fulfilled by** taking any courses offered by other study programs in SU upon approval from the Academic Advisor, approval from the Head of Study Program, and acknowledgement by the Dean of FET. **Please note that these courses will be recorded as additional credits and will not count toward CS degree requirements.** These courses can be found from the degree plan of each study program in SU, they are:
 - Faculty of Engineering and Technology:
 - Information System
 - Industrial Engineering
 - Mechanical Engineering
 - Visual Communication Design
 - Faculty of Business:
 - Accounting
 - Management

- Faculty of Education:
 - English Language Education

Remark:

*Each study program should prepare a list of courses that can be taken by students from other study programs (external students). Instead of using pre-requisite courses as the course requirement, these courses should apply pre-requisite skills/competencies as the course requirement for external students. The pre-requisite skills / competencies requirement should be added into the RPS of the associated courses. See **Table 13** for list of courses that can be taken at CS study program by students from other study programs.*

B. Maximum of 40 credits could be earned at different institutions as the following:

- Courses taken at different higher education institutions (subject to approval from the Academic Advisor, approval from the Head of Study Program, and acknowledgement by the Dean of FET)
- MBKM Internship activities at company or recognized institutions that can be converted into equivalent courses in SU

The combined credits from the two options above must not exceed 40 credits.

Credits that can be earned at different institutions are courses in semesters 3 – 8, which are listed in **Table 12**. For the Capstone 1 and 2 (*Tugas Akhir*), they can be taken in either a university or a company. In the case of taking Capstone in a company, the student should have two supervisors: one from the company and another one from CS.

Table 12. List of courses that could be taken at different Institutions

No	Course	Credits	Semester	Recommended Institution*
1	Calculus III	3	3	Keiser University
2	General Physics II + Lab	4	3	Keiser University
3	Bahasa Indonesia	2	3	ITB, UGM, PU
4	Kewarganegaraan	3	3	ITB, UGM, PU
5	Differential Equations	3	4	Keiser University
6	General Psychology	3	4 / 6	Keiser University
7	Macroeconomics	3	4 / 6	Keiser University
8	US History	3	4 / 6	Keiser University
9	Basic Circuits	5	5	UA
10	Microprocessor Organization + Lab	4	6	UA
11	Circuit Theory	3	6	UA
12	Internship	3	6 – 7	Company
13	Electronic Circuits	4	7	UA

14	Intro. To Communications	3	7	UA
15	Computer Architecture & Design	3	7	UA
16	Capstone 1	3	7	UA, ITB, UGM, PU, APU, UNITEN, Company
17	Knowledge System Engineering	3	8	UA
18	Fundamentals of Information and Network Security	3	8	UA
19	Computer Aided Logic Design	3	8	UA
20	Capstone II	3	8	UA, ITB, UGM, PU, APU, UNITEN, Company

* UA = University of Arizona, ITB = Institut Teknologi Bandung, UGM = Universitas Gadjah Mada, PU = President University, APU = Asia Pacific University, Malaysia, UNITEN = Universiti Tenaga Nasional, Malaysia

For the courses at other universities, CS Study Program will review the PLOs of the counterpart courses in partner universities. To apply for this scheme (and any MBKM scheme), students need to submit their application to CS study program at least 2 months prior to the start of the semester. In response to this scheme, CS study program will also prepare courses that can be taken by students from other universities / study program (see **Remark** and **Table 13**). The RPS of these courses should be extended to facilitate pre-requisite skills / competencies applicable to external students (see **Remark**).

Table 13. List of CS courses offered to students from other study programs in SU or other universities

No	Code	Course Name	Credits
1	COMP1401	Computer Programming	4
2	COMP2306	Operating Systems	3
3	COMP3309	Computer Networks	3
4	COMP3311	Software Engineering	3
5	COMP3315	Artificial Intelligence	3
6	STEM4302	Capstone I	3
7	STEM4303	Capstone II	3

Capstones 1 & 2 can be taken at partner university, or it can be taken in SU while the project itself is done at partner institution outside of SU.

Internship in CS study program can be done for the period of 2 - 6 months. The 2 months internship is held in Summer Semester between 6th and 7th semester. The 2 months internship activities are converted into STEM4301 – Internship, which is a 3 credits course in

CS Study Program / FET. The internship that is more than 2 months is recommended to be taken in:

- (1) Summer semester + early of 7th semester
- (2) After Capstone 2 defense and before the Judicium/graduation

When a student is accepted in an MBKM Internship program, which typically lasts for 6 months, the extra activities could be converted into equivalent course(s) offered at SU (in addition to the STEM4301). The conversion into equivalent courses in SU is subject to the assessment by CS study program. The 6 months MBKM Internship program is possible when students do their full 3 months internship in summer semester and continue for the next 3 months in the early 7th semester. Alternatively, the 6 months MBKM internship program would follow the schedule determined by DIKTI, and it might be scheduled in the student's 7th or 8th semester that might include Capstone 1 or 2. During this MBKM internship period, students should still be allowed by the company to attend academic activities.

Additional credits could also be earned when a student is admitted into MBKM internship program after the Capstone 2 defense and before the Judicium. However, the total credits from MBKM internship activities must be monitored so that the combined credits with the courses taken at different universities do not exceed 40 credits.

8. Management of Curriculum

Curriculum management is carried out by formulating the curriculum after the processes of evaluation and necessary revisions of the previous curriculum have been done. In its supervision, a review of the RPS of each course / subject is performed and its application is monitored. For example: the formulated CPL/SLO is reviewed for compliance with the relevant PLO by the curriculum team who understands the subject discussed in the meeting. The implementation of SLO is viewed from the assessment of SLO measurements, such as homework, quizzes, exams, presentations, final projects, etc. The implementation of the curriculum is reviewed by the measurement results a few times in a semester, which is usually discussed in the monthly CS study program meetings. The Learning Management System by Canvas platform is used for the monitoring of PLO achievement for every course, through Signature Assignment. At the end of every semester, the results are compiled and discussed among the lecturers in Computer Science Study Program.

Conclusion

The 2024 CS Study Program Curriculum is a result of revisions of many elements in the 2021 Curriculum. The changes are made after considering the input from stakeholders, and it is hoped that the new curriculum can help to ensure the learning outcomes and graduate profiles are achieved by our graduates. The new curriculum is also part of the efforts toward continuous improvements in CS study program. We believe that the new curriculum is not free from errors, and continuous curriculum review is still needed in the future.